THE DERIVED-CHOW KERNEL RESTRICTED TO V EVALUATES AS THE CLOSED-IMMERSION PUSHFORWARD FROM W

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 12114-12126 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 12128-12162, the complete author proof (33 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (both complete). Consumed by same-wave L16205.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma derived-chow (setup and property (5), in-source); Morphisms of Spaces, lemma finite-presentation-permanence; Derived Categories of Spaces, lemma cohomology-base-change.

Source reference. Original lemma 0CSJ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12013–12030. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05WT: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12162–12162. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08IN: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12162–12162. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let X/A, $x \in |X|$, and n, Z, z, V, E be as in Lemma 0CSJ. For any $K \in D_{QCoh}(\mathcal{O}_X)$ we have*

$$Rq_*(Lp^*K \otimes^{\mathbf{L}} E)|_V = R(W \to V)_*K|_W$$

where $p : X \times_A \mathbf{P}^n_A \to X$ and $q : X \times_A \mathbf{P}^n_A \to \mathbf{P}^n_A$ are the projections and where the morphism $W \to V$ is the finitely presented closed immersion $c|_W : W \to V$.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Since $W = c^{-1}(V)$ and since c is a closed immersion over V, we see that $c|_W$ is a closed immersion. It is of finite presentation because W and V are of finite presentation over A, see Morphisms of Spaces, Lemma 05WT. First we have

$$Rq_*(Lp^*K \otimes^{\mathbf{L}} E)|_V = Rq'_*\left((Lp^*K \otimes^{\mathbf{L}} E)|_{X \times_A V}\right)$$

where $q' : X \times_A V \to V$ is the projection because formation of total direct image commutes with localization. Denote $i = (b, c)|_W : W \to X \times_A V$ the given closed immersion. Then

$$Rq'_*\left((Lp^*K \otimes^{\mathbf{L}} E)|_{X \times_A V}\right) = Rq'_*(Lp^*K|_{X \times_A V} \otimes^{\mathbf{L}} i_*\mathcal{O}_W)$$

by property (5). Since i is a closed immersion we have $i_*\mathcal{O}_W = Ri_*\mathcal{O}_W$. Using Derived Categories of Spaces, Lemma 08IN we can rewrite this as

$$Rq'_*Ri_*Li^*Lp^*K|_{X \times_A V} = R(q' \circ i)_*Lb^*K|_W = R(W \to V)_*K|_W$$

which is what we want. (Note that restricting to W and derived pulling back via $W \to X$ is the same thing as W is étale over X.) □